NETAJI SUBHASH ENGINEERING COLLEGE

SUBJECT :- Computer Networks PCC-CS602(6596)

Topic Name :- "2) Flow and Error Control Protocols in Data Communication"

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2) write a report describing the operation and performance of flow and error control protocols, namely Stop-and wait ARQ, Go-Back-N ARQ, and selective Repeat ARQ, highlighting their advantages and limitations in practical network.

⇒ ◉ **Introduction :-** In data communication networks, reliable data transfer is achieved using flow control and error control mechanisms.

◉ **Flow control :-** Flow control ensures that a fast sender does not overwhelm a slow receiver. It regulates the rate of data transmission between sender and receiver.

◉ **Error control :-** Error control ensures that corrupted or lost frames are detected and retransmitted. It use :-

- Error detection techniques (CRC, checksum)
- Acknowledgment (ACK/NAK)
- Time-out mechanism.

- Automatic Repeat request (ARQ) protocols combine both flow and error control.

The three main ARQ protocols are.

i) stop-and-wait ARQ
ii) Go-Back-N ARQ
iii) selective Repeat ARQ.

Each protocol improves efficiency and reliability in different network conditions.

Stop-and-wait ARQ:-

- **Operation:-**
  - The sender transmits one frame.
  - If ACK is received → send next frame.
  - Uses 1-bit sequence number (0 and 1).
  - window size = 1.

- **Flow control:-** only one outstanding frame is allowed.

  This prevents receiver overflow.

- **Error control:-**
  - Error detection using CRC/checksum.
  - Timeout-based retransmission.

- **Performance:-**
  - **Efficiency:-** $\frac{T_f}{T_f + 2T_p}$

- **where:-**
  - T_f = Frame transmission time.
  - T_p = propagation delay.

  In terms of $a = \frac{T_p}{T_f}$:

  - **Efficiency** $= \frac{1}{1+2a}$

Advantages:-
- very simple.
- Easy implementation.
- Low memory requirement.

Go-Back-N (GBN) ARQ :-

operation
- sender can transmit multiple frames before receiving Ack.
- using sliding window mechanism.
- Frames numbered modulo 2^m.
- sender window size:

$$W \leq 2^m - 1$$

- Reciver window size = 1.
- If frame k is lost → all subsequent frames discarded.
- sender retransmits from k.

Flow control:-
- sliding window controls number of outstanding frames.
- prevents receiver overflow.

Error control:-
- cumulative acknowledgment.
- Time out causes retransmission of all unacknowledged frames.

Performance:-

$$\text{Efficiency (no error)} = \frac{w}{1+2a}$$

where:-

- w = window size
- $a = \frac{T_p}{T_f}$

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- $a = \frac{T_p}{T_f}$

Efficiency decreases when error rate increases.

Advantages:-

- Higher efficiency than stop-and wait.
- Better bandwidth utilization.
- Suitable for moderate error networks.

Limitations:-

- Retransmits many correct frames.
- wates bandwidth when errors occur.
- Receiver window size limited to 1.

Selective Repeat (SR) ARQ :-

Operation :-

- Receiver accepts out-of-order frames.
- Only erroneous frames are retransmitted.
- Both sender and receiver use sliding window.

Window size constraint :-

$$w \leq 2^{m-1}$$

- This avoids sequence number ambiguity.

Flow control :-

- Sender window size = w.
- Receiver window size = w.
- Buffering required at receiver.

Error control :-

- Selective retransmission.
- Out-of-order buffering.

Performance :-

- Efficiency (no error) :- $\frac{w}{1+2a}$

With errors, performance remains high because only lost frames are retransmited.

Advantages:-

- Highest efficiency.
- Minimal retransmission overhead.

Limitations:-

- complex implementation.
- High buffer requirement.

comparative Analysis and conclusion:-

- comparison table:-

Feature	Stop-and-wait	Go-Back-N	Selective Repeat
windowsize	1	$\leq 2^m - 1$	$\leq 2^m - 1$
Receiver window	1	1	W
efficiency	Low	medium	High
Retransmission	one frame	multiple frames	only lost frames
complexity	low	Medium	High
Buffer Requirement	Low	Moderate	High.

pratical Network use:-

- Stop-and-wait:- used in simple, low-speed links.
- Go-Back-N:- used in networks with low error rates.
- selective Repeat:- used in modern high-speed and long-delay networks.

Conclusion:-

Flow and error control protocols ensure reliable data transmission in networks. Stop-and-wait is simple but has low efficiency. Go-Back-N improves bandwidth utilization but may waste resources during errors.

Selective Repeat offers the highest efficiency by retransmitting only lost frames, though it is more complex. The choice of protocol depends on delay, bandwidth, error rate, and system complexity.